

18

Reinsurance

Syllabus objectives

- 1.1 Loss distributions, with and without risk sharing
 - 1.1.2 Explain the concepts of excesses (deductibles), and retention limits.
 - 1.1.3 Describe the operation of simple forms of proportional and excess of loss reinsurance.
 - 1.1.4 Derive the distribution and corresponding moments of the claim amounts paid by the insurer and the reinsurer in the presence of excesses (deductibles) and reinsurance.
 - 1.1.5 Estimate the parameters of a failure time or loss distribution when the data is complete, or when it is incomplete, using maximum likelihood and the method of moments.

0 Introduction

The claims on an insurance company must be met in full, but, to protect itself from large claims, the company itself may take out an insurance policy; such a policy is called a reinsurance policy. For the purposes of this chapter, it will be assumed that the reinsurance contract is one of two very simple types: individual excess of loss reinsurance or proportional reinsurance.

0.1 Proportional reinsurance

Under a proportional reinsurance arrangement, the *direct writer* (ie the original insurance company) and the reinsurer share the cost of all claims for each risk. For example, for a particular building insured against fire, the direct writer might retain 75% of the premium and will be liable to pay 75% of all claims, large or small. The direct writer must pay a premium to effect this reinsurance. The direct writer is sometimes referred to as the *direct insurer* or even just the *insurer*.

Proportional reinsurance operates in two forms:

1. With *quota share* reinsurance, the proportions are the same for all risks.
2. With *surplus* reinsurance, the proportions can vary from one risk to the next.

In this course we will focus on quota share reinsurance.

0.2 Non-proportional reinsurance

Under a non-proportional reinsurance arrangement, the direct writer pays a fixed premium to the reinsurer. The reinsurer will only be required to make payments where part of the claim amount falls in a particular reinsurance *layer* (eg between £1m and £5m). The layer will be defined by a lower limit, the *retention limit* (eg £1m), and an upper limit (eg £5m or infinity if the cover is *unlimited*). Usually, most claims are paid in full by the direct writer.

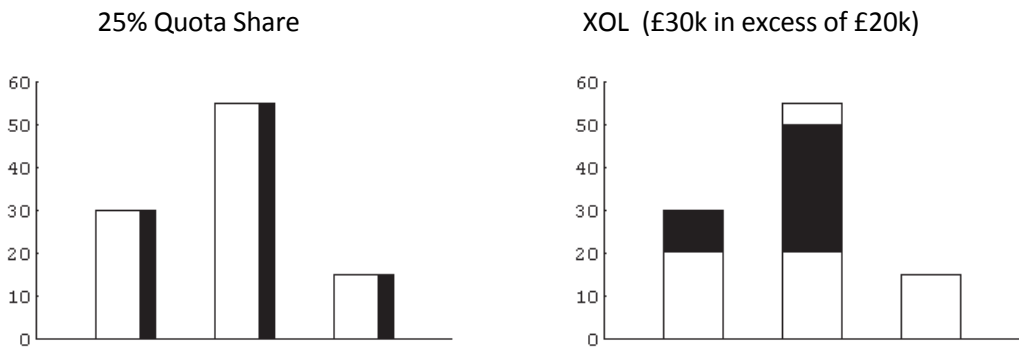
We will mention two forms of non-proportional reinsurance here:

1. With *individual excess of loss* (XOL) reinsurance, the reinsurer will be required to make a payment when the claim amount for any individual claim exceeds a specified *excess point* or *retention*. For example, the reinsurer might agree to pay the excess when any claim from a motor policy exceeds £50,000, but with an upper limit of £2 million.
2. With *stop loss* reinsurance, the reinsurer will be required to make payments if the total claim amount for a specified group of policies exceeds a specified amount (which may be expressed as a percentage of the gross premium). We will look at this in [Chapter 20](#).

The diagram below shows how much the direct writer and the reinsurer would pay when there are claims for £30,000, £55,000 and £15,000:

- (a) under a 25% quota share arrangement, and
- (b) under an individual XOL arrangement with a reinsurance layer of £30,000 in excess of £20,000.

The parts of each claim paid by the reinsurer are shown in black.



1 Reinsurance arrangements

The actual amount that the direct insurer ends up paying after allowing for payments under the reinsurance arrangements is called the *net claim amount*. The actual premium that the direct writer gets to keep after making any payments for reinsurance is the *insurer's net premium income*. The original amounts without adjustment for reinsurance are referred to as the *gross claim amount* and the *insurer's gross premium income*.

In this chapter we will use the following notation.



Notation

X is the gross claim amount random variable

Y is the net claim amount, *ie* the amount of the claim paid by the insurer in respect of a single claim (after receiving the reinsurance recovery)

Z is the amount paid by the reinsurer in respect of a single claim.

For a given reinsurance arrangement, we can express the random variables Y and Z in terms of X .

For example, suppose that a reinsurer has agreed to make the following payments in respect of individual claims incurred by a direct insurer:

- nothing, if the claim is less than £5,000
- the full amount reduced by £5,000, if the claim is between £5,000 and £10,000
- half the full amount, if the claim is between £10,000 and £20,000
- £10,000, if the claim exceeds £20,000.

Then:

$$Z = \begin{cases} 0 & \text{if } X \leq 5,000 \\ X - 5,000 & \text{if } 5,000 < X \leq 10,000 \\ X / 2 & \text{if } 10,000 < X \leq 20,000 \\ 10,000 & \text{if } X > 20,000 \end{cases}$$

and:

$$Y = \begin{cases} X & \text{if } X \leq 5,000 \\ 5,000 & \text{if } 5,000 < X \leq 10,000 \\ X / 2 & \text{if } 10,000 < X \leq 20,000 \\ X - 10,000 & \text{if } X > 20,000 \end{cases}$$

Note that $Y + Z = X$.

We are now in a position to consider the statistical calculations relating to reinsurance arrangements.

1.1 Excess of loss reinsurance

In excess of loss reinsurance, the insurer will pay any claim in full up to an amount M , the retention level; any amount above M will be borne by the reinsurer.

The excess of loss reinsurance arrangement can be written in the following way: if the claim is for amount X , then the insurer will pay Y where:

$$Y = X \quad \text{if } X \leq M$$

$$Y = M \quad \text{if } X > M$$

The reinsurer pays the amount $Z = X - Y$.



Question

Write down an expression for Y if only a layer between M and $2M$ is reinsured.

Solution

$$Y = \begin{cases} X & \text{if } X \leq M \\ M & \text{if } M < X \leq 2M \\ X - M & \text{if } X > 2M \end{cases}$$

The insurer's liability is affected in two obvious ways by reinsurance:

- (i) the mean amount paid is reduced;
- (ii) the variance of the amount paid is reduced.

Both these conclusions are simple consequences of the fact that excess of loss reinsurance puts an upper limit on large claims.

The mean amounts paid by the insurer and the reinsurer under excess of loss reinsurance can now be obtained. Observe that the mean amount paid by the insurer without reinsurance is:

$$E(X) = \int_0^{\infty} x f(x) dx \quad (18.1)$$

where $f(x)$ is the PDF of the claim amount X . With a retention level of M the mean amount paid by the insurer becomes:

$$E(Y) = \int_0^M x f(x) dx + M P(X > M) \quad (18.2)$$

This is because:

$$E(Y) = \int_0^M x f(x) dx + \int_M^{\infty} M f(x) dx = \int_0^M x f(x) dx + M \int_M^{\infty} f(x) dx$$

and:

$$\int_M^{\infty} f(x) dx = P(X > M)$$

We can calculate $E(Y^2)$ in a similar way:

$$E(Y^2) = \int_0^M x^2 f(x) dx + \int_M^{\infty} M^2 f(x) dx = \int_0^M x^2 f(x) dx + M^2 P(X > M)$$

$$\text{Then } \text{var}(Y) = E(Y^2) - [E(Y)]^2.$$

More generally, the moment generating function of Y , the amount paid by the insurer, is:

$$M_Y(t) = E(e^{tY}) = \int_0^M e^{tx} f(x) dx + e^{tM} P(X > M)$$

Here we are using the formula for the expected value of a function of a continuous random variable:

$$E(h(X)) = \int_x h(x) f(x) dx$$

with:

$$h(X) = \begin{cases} e^{tX} & \text{if } X \leq M \\ e^{tM} & \text{if } X > M \end{cases}$$



Question

Suppose that claim amounts are uniformly distributed over the interval $(0, 500)$. The insurer effects individual excess of loss reinsurance with a retention limit of 375.

Calculate the expected amounts paid by the insurer and the reinsurer in respect of a single claim.

Solution

Since $X \sim U(0, 500)$, the expected gross claim amount is:

$$E(X) = \frac{500}{2} = 250$$

The expected amount paid by the insurer is:

$$\begin{aligned}
 E(Y) &= \int_0^{375} x f(x) dx + 375 P(X > 375) \\
 &= \int_0^{375} \frac{x}{500} dx + 375 [1 - F(375)] \\
 &= \left[\frac{x^2}{1,000} \right]_0^{375} + 375 \left[1 - \frac{375}{500} \right] \\
 &= 140.625 + 93.75 \\
 &= 234.375
 \end{aligned}$$

Also, since $Y + Z = X$:

$$E(Z) = E(X) - E(Y) = 250 - 234.375 = 15.625$$

Under excess of loss reinsurance, the reinsurer will pay Z where:

$$Z = \begin{cases} 0 & \text{if } X \leq M \\ X - M & \text{if } X > M \end{cases}$$

The mean amount paid by the reinsurer is:

$$E(Z) = \int_M^{\infty} (x - M) f(x) dx \quad (18.3)$$

Similarly, we can calculate $E(Z^2)$ using:

$$E(Z^2) = \int_M^{\infty} (x - M)^2 f(x) dx$$

Then $\text{var}(Z) = E(Z^2) - [E(Z)]^2$.

More generally, the moment generating function of Z is:

$$\begin{aligned}
 M_Z(t) &= E(e^{tZ}) \\
 &= \int_0^M e^{t \cdot 0} f(x) dx + \int_M^{\infty} e^{t(x-M)} f(x) dx \\
 &= \int_0^M f(x) dx + \int_M^{\infty} e^{t(x-M)} f(x) dx \\
 &= P(X \leq M) + \int_M^{\infty} e^{t(x-M)} f(x) dx
 \end{aligned}$$

We can use R to simulate gross claim amounts and hence the amounts paid by the insurer and reinsurer for any given retention limit.



Suppose claims (in £'s) have an exponential distribution with parameter $\lambda = 0.0005$. The R code for simulating 10,000 claims, x , is given by:

```
x <- rexp(10000, rate=0.0005)
```

We can then obtain the claims paid by the insurer, y , and the reinsurer, z with retention M using:

```
y <- pmin(x, M)
z <- pmax(0, x-M)
```

We can then obtain their means and variances using the R functions `mean` and `var`.

We can use these vectors to estimate probabilities. For example, to estimate the probability that the insurer pays less than £1,000 we would use:

```
length(y[y<1000])/length(y)
```

Similarly we could estimate the claim size for a given percentile. For example, to estimate the claim size corresponding to the 90th percentile of the insurer's claims we would use:

```
quantile(y, 0.9)
```

1.2 The reinsurer's conditional claims distribution

Now consider reinsurance (once again) from the point of view of the reinsurer. The reinsurer may have a record only of claims that are greater than M . If a claim is for less than M the reinsurer may not even know a claim has occurred. The reinsurer thus has the problem of estimating the underlying claims distribution when only those claims greater than M are observed. The statistical terminology is to say that the reinsurer observes claims from a truncated distribution.

In this case the values observed by the reinsurer relate to a conditional distribution, since the numbers are conditional on the original claim amount exceeding the retention limit.

Let W be the random variable with this truncated distribution. Then:

$$W = X - M \mid X > M$$

This can also be expressed as follows:

$$W = Z \mid Z > 0$$

Suppose that the underlying claim amounts have PDF $f(x)$ and CDF $F(x)$. Suppose that the reinsurer is only informed of claims greater than the retention M and has a record of $w = x - M$. What is the PDF $g(w)$ of the amount, w , paid by the reinsurer?

The argument goes as follows:

$$\begin{aligned}
 P(W < w) &= P(X < w + M \mid X > M) \\
 &= \frac{P(X < w + M \text{ and } X > M)}{P(X > M)} \\
 &= \frac{P(M < X < w + M)}{P(X > M)} \\
 &= \int_M^{w+M} \frac{f(x)}{1 - F(M)} dx \\
 &= \frac{F(w + M) - F(M)}{1 - F(M)}
 \end{aligned}$$

This derivation also uses the result:

$$P(a < X < b) = \int_a^b f(x) dx = F(b) - F(a)$$

Differentiating with respect to w , the PDF of the reinsurer's claims is:

$$g(w) = \frac{f(w + M)}{1 - F(M)}, \quad w > 0 \quad (18.4)$$

This is just the original PDF evaluated at the gross amount $w + M$, divided by the probability that the claim exceeds M .

The PDF of W may be denoted by $f_W(w)$ rather than $g(w)$. With this notation, the result can be stated as follows:



PDF of the reinsurer's conditional claim amount random variable

If $W = X - M \mid X > M$, then:

$$f_W(w) = \frac{f_X(w + M)}{1 - F_X(M)} = \frac{f_X(w + M)}{P(X > M)}$$



Question

Using the notation above, determine the distribution of W if:

- (a) $X \sim \text{Exp}(\lambda)$
- (b) $X \sim \text{Pa}(\alpha, \lambda)$

Solution

(a) Exponential

If $X \sim \text{Exp}(\lambda)$, then $f_X(x) = \lambda e^{-\lambda x}$ and $F_X(x) = 1 - e^{-\lambda x}$. So:

$$f_W(w) = \frac{f_X(w+M)}{1-F_X(M)} = \frac{\lambda e^{-\lambda(w+M)}}{e^{-\lambda M}} = \lambda e^{-\lambda w}, \quad w > 0$$

This is the PDF of $\text{Exp}(\lambda)$. So $W \sim \text{Exp}(\lambda)$, the same as the original claims distribution. This illustrates the memoryless property of the exponential distribution.

(b) Pareto

If $X \sim \text{Pa}(\alpha, \lambda)$, then $f_X(x) = \frac{\alpha \lambda^\alpha}{(\lambda + x)^{\alpha+1}}$ and $F_X(x) = 1 - \left(\frac{\lambda}{\lambda + x}\right)^\alpha$. So:

$$f_W(w) = \frac{\alpha \lambda^\alpha / (\lambda + w + M)^{\alpha+1}}{\lambda^\alpha / (\lambda + M)^\alpha} = \frac{\alpha (\lambda + M)^\alpha}{(\lambda + M + w)^{\alpha+1}}, \quad w > 0$$

This is the PDF of $\text{Pa}(\alpha, \lambda + M)$. So $W \sim \text{Pa}(\alpha, \lambda + M)$.

We can now calculate the expected value of W . Using the PDF of $W = X - M \mid X > M$, we have:

$$E(W) = \int_0^\infty w f_W(w) dw = \frac{\int_0^\infty w f_X(w+M) dw}{1-F_X(M)} = \frac{\int_M^\infty (x-M) f_X(x) dx}{1-F_X(M)} = \frac{E(Z)}{P(X > M)} = \frac{E(Z)}{P(Z > 0)}$$

So the reinsurer's expected claim payment on a claim in which it is involved is just the reinsurer's expected claim payment (on all claims), $E(Z)$, divided by the probability that the claim involves the reinsurer.



If z is the vector of the reinsurer's claims:

```
z <- pmax(0, x-M)
```

Then we can obtain the truncated distribution, w , using:

```
w <- z[z>0]
```

We can then calculate moments, probabilities and quantiles as before.

1.3 Proportional reinsurance

In proportional reinsurance the insurer pays a fixed proportion of the claim, whatever the size of the claim. Using the same notation as above, the proportional reinsurance arrangement can be written as follows: if the claim is for an amount X then the company will pay Y where:

$$Y = \alpha X \quad 0 < \alpha < 1$$

The parameter α is known as the retained proportion or retention level; note that the term retention level is used in both excess of loss and proportional reinsurance though it means different things.

Since $Y + Z = X$, we must have $Z = (1 - \alpha)X$. The mean and variance of Y and Z are calculated as follows:

$$E(Y) = \alpha E(X) \quad E(Z) = (1 - \alpha)E(X)$$

$$\text{var}(Y) = \alpha^2 \text{var}(X) \quad \text{var}(Z) = (1 - \alpha)^2 \text{var}(X)$$



Question

Claims from a particular portfolio have a generalised Pareto distribution with parameters $\alpha = 6$, $\lambda = 200$ and $k = 4$. A proportional reinsurance arrangement is in force with a retained proportion of 80%.

Calculate the mean and variance of the amount paid by the insurer and the reinsurer in respect of a single claim.

Solution

Using X to represent the individual claim amount random variable and the formulae for the mean and variance of a three-parameter Pareto random variable (from page 15 of the *Tables*), we have:

$$E(X) = \frac{k\lambda}{\alpha - 1} = \frac{4 \times 200}{6 - 1} = \frac{800}{5} = 160$$

and:

$$\text{var}(X) = \frac{k(k + \alpha - 1)\lambda^2}{(\alpha - 1)^2(\alpha - 2)} = \frac{4(4 + 6 - 1) \times 200^2}{(6 - 1)^2(6 - 2)} = \frac{1,440,000}{100} = 14,400$$

The amount paid by the insurer is $Y = 0.8X$. So:

$$E(Y) = 0.8 \times 160 = 128$$

and:

$$\text{var}(Y) = 0.8^2 \times 14,400 = 9,216$$

The amount paid by the reinsurer is $Z = 0.2X$. So:

$$E(Z) = 0.2 \times 160 = 32$$

and:

$$\text{var}(Z) = 0.2^2 \times 14,400 = 576$$

As the amount paid by the insurer on a claim X is $Y = \alpha X$ and the amount paid by the reinsurer is $Z = (1 - \alpha)X$, the distribution of both of these amounts can be found by a simple change of variable.



Question

Claims from a particular portfolio have an exponential distribution with mean 1,000. The insurer takes out proportional reinsurance with a retained proportion of 0.9.

Determine the distribution of the insurer's net claim amount random variable.

Solution

We know that X is exponential with mean 1,000, so the exponential parameter is $\frac{1}{1,000}$.

From page 11 of the *Tables*, the MGF of X is:

$$M_X(t) = (1 - 1,000t)^{-1}, \quad t < \frac{1}{1,000}$$

Since $Y = 0.9X$, the MGF of Y is:

$$M_Y(t) = E(e^{tY}) = E(e^{0.9tX}) = M_X(0.9t) = (1 - 1,000 \times 0.9t)^{-1} = (1 - 900t)^{-1}, \quad t < \frac{1}{900}$$

This is the MGF of the exponential distribution with mean 900. By the uniqueness property of MGFs, it follows that the distribution of the insurer's net claim amount random variable is exponential with mean 900.



The payments of the insurer, y , and the reinsurer, z , with retained proportion a would be:

$$\begin{aligned} y &\leftarrow a * x \\ z &\leftarrow (1-a) * x \end{aligned}$$

We can then calculate moments, probabilities and quantiles as before.

2 Normal and lognormal distributions

There are useful integral formulae that simplify reinsurance calculations when working with normal and lognormal distributions.

2.1 Normal distribution



Truncated mean of the normal distribution

If $X \sim N(\mu, \sigma^2)$, then:

$$\int_L^U x f_X(x) dx = \mu [\Phi(U') - \Phi(L')] - \sigma [\phi(U') - \phi(L')]$$

where:

$$L' = \frac{L - \mu}{\sigma}$$

$$U' = \frac{U - \mu}{\sigma}$$

$\phi(z)$ is the PDF of the standard normal distribution

$\Phi(z)$ is the CDF of the standard normal distribution.

This result is given on page 18 of the *Tables*. It is proved as follows.

Using the formula for $f_X(x)$ and the substitution $z = \frac{x - \mu}{\sigma}$:

$$\begin{aligned} \int_L^U x f_X(x) dx &= \int_L^U x \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \\ &= \int_{L'}^{U'} (\mu + \sigma z) \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz \\ &= \mu \int_{L'}^{U'} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz + \sigma \int_{L'}^{U'} z \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz \end{aligned}$$

Now, since $\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$ is the PDF of $N(0,1)$:

$$\int_{L'}^{U'} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz = P(L' < N(0,1) < U')$$

So:

$$\begin{aligned}
 \int_L^U x f_X(x) dx &= \mu P(L' < N(0,1) < U') + \sigma \int_{L'}^{U'} z \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz \\
 &= \mu P(L' < N(0,1) < U') + \sigma \left[-\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} \right]_{L'}^{U'} \\
 &= \mu [\Phi(U') - \Phi(L')] - \sigma [\phi(U') - \phi(L')] \\
 &= \mu \left[\Phi\left(\frac{U-\mu}{\sigma}\right) - \Phi\left(\frac{L-\mu}{\sigma}\right) \right] - \sigma \left[\phi\left(\frac{U-\mu}{\sigma}\right) - \phi\left(\frac{L-\mu}{\sigma}\right) \right]
 \end{aligned}$$

When $L = -\infty$ or $U = \infty$, these formulae can be simplified because:

$$\phi(-\infty) = \phi(\infty) = 0, \quad \Phi(-\infty) = 0, \quad \Phi(\infty) = 1$$



Question

Claims from a particular portfolio are normally distributed with mean 800 and standard deviation 100. An individual excess of loss arrangement with retention limit is 860 is in place.

Calculate the insurer's mean claim payment net of reinsurance.

Solution

The insurer's mean claim payment is:

$$E(Y) = \int_0^{860} x f_X(x) dx + 860 P(X > 860)$$

where $X \sim N(800, 100^2)$.

Using the formula for the truncated mean of a normal random variable:

$$\begin{aligned}
 \int_0^{860} x f_X(x) dx &= 800 \left[\Phi\left(\frac{860-800}{100}\right) - \Phi\left(\frac{0-800}{100}\right) \right] \\
 &\quad - 100 \left[\phi\left(\frac{860-800}{100}\right) - \phi\left(\frac{0-800}{100}\right) \right] \\
 &= 800 [\Phi(0.6) - \Phi(-8)] - 100 [\phi(0.6) - \phi(-8)]
 \end{aligned}$$

From pages 160 and 161 of the *Tables*:

$$\Phi(0.6) = 0.72575$$

$$\Phi(-8) = 1 - \Phi(8) \approx 0$$

Also, using the formula for the PDF of the standard normal distribution from page 10 of the *Tables*:

$$\phi(0.6) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \times 0.6^2} = 0.33322$$

$$\phi(-8) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \times (-8)^2} \approx 0$$

So:

$$\int_0^{860} x f_X(x) dx \approx 800[0.72575 - 0] - 100[0.33322 - 0] = 547.278$$

The second term in the expression for $E(Y)$ is:

$$\begin{aligned} 860 P(X > 860) &= 860 \left[1 - \Phi\left(\frac{860 - 800}{100}\right) \right] \\ &= 860[1 - \Phi(0.6)] \\ &= 860[1 - 0.72575] \\ &= 235.855 \end{aligned}$$

Hence:

$$E(Y) \approx 547.278 + 235.855 = 783.13$$

2.2 Lognormal distribution



Truncated moments of the lognormal distribution

If $X \sim \log N(\mu, \sigma^2)$, then:

$$\int_L^U x^k f_X(x) dx = e^{k\mu + \frac{1}{2}k^2\sigma^2} [\Phi(U_k) - \Phi(L_k)]$$

where:

$$L_k = \frac{\ln L - \mu}{\sigma} - k\sigma$$

$$U_k = \frac{\ln U - \mu}{\sigma} - k\sigma$$

$\Phi(z)$ is the CDF of the standard normal distribution.

This result is also given on page 18 of the *Tables*. It is proved as follows.

Using the formula for the PDF of the lognormal distribution from page 14 of the *Tables*, we have:

$$\int_L^U x^k f_X(x) dx = \int_L^U x^k \frac{1}{\sigma\sqrt{2\pi}} \frac{1}{x} e^{-\frac{1}{2}\left(\frac{\ln x - \mu}{\sigma}\right)^2} dx$$

Making the substitution $t = \frac{\ln x - \mu}{\sigma} - k\sigma$ gives:

$$\begin{aligned} \int_L^U x^k f_X(x) dx &= \int_{L_k}^{U_k} e^{k(\mu + \sigma t + k\sigma^2)} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(t + k\sigma)^2} dt \\ &= \int_{L_k}^{U_k} \frac{1}{\sqrt{2\pi}} e^{k\mu + k\sigma t + k^2\sigma^2} e^{-\frac{1}{2}t^2 - k\sigma t - \frac{1}{2}k^2\sigma^2} dt \\ &= e^{k\mu + \frac{1}{2}k^2\sigma^2} \int_{L_k}^{U_k} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}t^2} dt \\ &= e^{k\mu + \frac{1}{2}k^2\sigma^2} [\Phi(U_k) - \Phi(L_k)] \end{aligned}$$

When $L = 0$ or $U = \infty$, these formulae can be simplified using the facts that:

$$\Phi(-\infty) = 0, \Phi(0) = \frac{1}{2}, \Phi(\infty) = 1$$

By setting $k = 1$ in the truncated moments formula, we can calculate the insurer's expected claim payment under excess of loss reinsurance when the original claims follow a lognormal distribution.



Question

An insurer is considering taking out one of the following reinsurance treaties:

Treaty 1: Proportional reinsurance with a retained proportion of 0.75

Treaty 2: Individual excess of loss cover with a retention limit of £25,000

The claims distribution is lognormal with parameters $\mu = 8.5$ and $\sigma^2 = 0.64$.

Calculate the insurer's expected net claim payments in the following cases:

- (a) without either treaty
- (b) with Treaty 1 only
- (c) with Treaty 2 only.

Solution

(a) No reinsurance

Without either treaty, the insurer pays the full amount of each loss. So:

$$E(Y) = E(X) = e^{\mu + \frac{1}{2}\sigma^2} = e^{8.5 + \frac{1}{2} \times 0.64} = \text{£}6,768$$

(b) Treaty 1

Under Treaty 1, the insurer pays 75% of each loss. So $Y = 0.75X$ and:

$$E(Y) = 0.75E(X) = 0.75 \times 6,768 = \text{£}5,076$$

(c) Treaty 2

Under Treaty 2, the insurer pays the first £25,000 of each loss. So:

$$Y = \begin{cases} X & \text{if } X \leq 25,000 \\ 25,000 & \text{if } X > 25,000 \end{cases}$$

and:

$$E(Y) = \int_0^{25,000} x f_X(x) dx + 25,000P(X > 25,000)$$

Using the truncated moments formula with $k = 1$ gives:

$$\int_0^{25,000} x f_X(x) dx = e^{\mu + \frac{1}{2}\sigma^2} [\Phi(U_1) - \Phi(L_1)]$$

where:

$$U_1 = \frac{\ln 25,000 - \mu}{\sigma} = \frac{\ln 25,000 - 8.5}{\sqrt{0.64}} = 1.23329$$

$$\begin{aligned} \Phi(U_1) &\approx (1 - 0.329)\Phi(1.23) + 0.329\Phi(1.24) \\ &= 0.671 \times 0.89065 + 0.329 \times 0.89251 \\ &= 0.89126 \end{aligned}$$

and:

$$\Phi(L_1) = \Phi\left(\frac{\ln L - \mu}{\sigma}\right) = 0 \quad \text{since } \ln L \rightarrow -\infty \text{ as } L \rightarrow 0$$

So:

$$\int_0^{25,000} x f_X(x) dx = e^{8.5 + \frac{1}{2} \times 0.64} \times 0.89126 = 6,032.30$$

The second term in the expression for $E(Y)$ is:

$$\begin{aligned} 25,000P(X > 25,000) &= 25,000 \left[1 - \Phi \left(\frac{\ln 25,000 - 8.5}{\sqrt{0.64}} \right) \right] \\ &= 25,000 [1 - \Phi(2.03329)] \end{aligned}$$

Interpolating gives:

$$\begin{aligned} \Phi(2.03329) &\approx (1 - 0.329)\Phi(2.03) + 0.329\Phi(2.04) \\ &= 0.671 \times 0.97882 + 0.329 \times 0.97932 \\ &= 0.97898 \end{aligned}$$

So:

$$E(Y) \approx 6,032.30 + 25,000 \times (1 - 0.97898) = \text{£}6,558$$

Setting $k = 2$ in the truncated moments formula, we can calculate the second non-central moment of the insurer's claim payment. We now extend the previous question to calculate the standard deviation of the insurer's net claim amount.



Question

An insurer is considering taking out one of the following reinsurance treaties:

Treaty 1: Proportional reinsurance with a retained proportion of 0.75

Treaty 2: Individual excess of loss cover with a retention limit of £25,000

The claims distribution is lognormal with parameters $\mu = 8.5$ and $\sigma^2 = 0.64$.

Calculate the standard deviation of the insurer's net claim payments in the following cases:

- (a) without either treaty
- (b) with Treaty 1 only
- (c) with Treaty 2 only.

Solution

(a) No reinsurance

Without either treaty, the variance is:

$$\text{var}(Y) = \text{var}(X) = e^{2\mu + \sigma^2} (e^{\sigma^2} - 1) = e^{2(8.5) + 0.64} (e^{0.64} - 1) = 41,067,256.6$$

So the standard deviation is £6,408.

(b) Treaty 1

Under Treaty 1:

$$\text{var}(Y) = 0.75^2 \text{var}(X)$$

So the standard deviation is $0.75 \times 6,408 = £4,806$.

(c) Treaty 2

Under Treaty 2:

$$E(Y^2) = \int_0^M x^2 f_X(x) dx + \int_M^\infty M^2 f_X(x) dx$$

Using the truncated moments formula with $k = 2$ gives:

$$\int_0^{25,000} x^2 f_X(x) dx = e^{2\mu + 2\sigma^2} [\Phi(U_2) - \Phi(L_2)]$$

where:

$$U_2 = \frac{\ln U - \mu}{\sigma} - 2\sigma = \frac{\ln 25,000 - 8.5}{\sqrt{0.64}} - 2\sqrt{0.64} = 0.43329$$

$$\begin{aligned} \Phi(U_2) &\approx (1 - 0.329)\Phi(0.43) + 0.329\Phi(0.44) \\ &= 0.671 \times 0.66640 + 0.329 \times 0.67003 \\ &= 0.66759 \end{aligned}$$

and:

$$\Phi(L_2) = \Phi\left(\frac{\ln L - \mu}{\sigma} - 2\sigma\right) = 0 \quad \text{since } \ln L \rightarrow -\infty \text{ as } L \rightarrow 0$$

So:

$$\int_0^{25,000} x^2 f_X(x) dx = e^{2 \times 8.5 + 2 \times 0.64} \times 0.66759 = 57,998,646$$

$$E(Y^2) = 57,998,646 + 25,000^2 \times (1 - 0.97898) = 71,130,942$$

$$\text{var}(Y) = 71,130,942 - 6,558^2 = 5,304^2$$

and the standard deviation is £5,304. (The calculations are very sensitive to rounding, so we have used Excel to obtain accurate values.)

Our calculations have shown that reinsurance reduces both the mean and variance of the insurer's claim payments, as expected.

3 Inflation

The examples we have considered so far have assumed that claim distributions don't change over time (or at least that we are looking at a sufficiently short time period for us to be able to make this assumption).

In practice claims are likely to increase because of inflation, at least in the longer term. A claim distribution that is suitable for modelling claim amounts in one year may well not be suitable a year or two later. We need to adjust our claim distributions to allow for inflation.

In this section we will look at how claims inflation affects reinsurance arrangements. It is easy to deal with claims inflation in the proportional reinsurance situation.



Question

Claims from a portfolio of policies are believed to follow an $\text{Exp}(\lambda)$ distribution. A proportional reinsurance arrangement with a retained proportion α is in force.

- (i) Give an expression for the insurer's expected claim payment.
- (ii) Next year, claim amounts are expected to increase by an inflationary factor of k . Derive an expression for the insurer's expected claim payment next year.

Solution

- (i) ***Expected claim payment this year***

We know that $X \sim \text{Exp}(\lambda)$ and $Y = \alpha X$. So:

$$E(Y) = E(\alpha X) = \alpha E(X) = \frac{\alpha}{\lambda}$$

- (ii) ***Expected claim payment next year***

Next year, the gross claim amount random variable is kX and the insurer's net claim payment is αkX . So the insurer's expected claim payment is:

$$E(\alpha kX) = \alpha k E(X) = \frac{\alpha k}{\lambda}$$

With excess of loss reinsurance, inflation can cause a problem. Suppose that the claims X are inflated by a factor of k but the retention M remains fixed. What effect does this have on the arrangement?

The amount claimed is kX , and the amount paid by the insurer, Y , is:

$$Y = kX \quad \text{if } kX \leq M$$

$$Y = M \quad \text{if } kX > M$$

In other words:

$$Y = \begin{cases} kX & \text{if } X \leq \frac{M}{k} \\ M & \text{if } X > \frac{M}{k} \end{cases}$$

The mean amount paid by the insurer is:

$$E(Y) = \int_0^{M/k} kx f(x) dx + MP(X > M/k) \quad (18.5)$$

For example, if $X \sim \text{Exp}(\lambda)$, then:

$$E(Y) = \int_0^{M/k} kx \lambda e^{-\lambda x} dx + MP(X > M/k)$$

Integrating by parts:

$$\begin{aligned} E(Y) &= \left[-kx e^{-\lambda x} \right]_0^{M/k} + \int_0^{M/k} k e^{-\lambda x} dx + MP(X > M/k) \\ &= -Me^{-\lambda M/k} + \int_0^{M/k} k e^{-\lambda x} dx + Me^{-\lambda M/k} \end{aligned}$$

The first and last terms in the line above cancel to give:

$$E(Y) = \int_0^{M/k} k e^{-\lambda x} dx = \left[-\frac{k}{\lambda} e^{-\lambda x} \right]_0^{M/k} = \frac{k}{\lambda} (1 - e^{-\lambda M/k})$$

One important general point that can be made is that the new mean claim amount paid by the insurer is not k times the mean claim amount paid by the insurer without inflation.

The insurer's mean claim amount will inflate by less than k . We can see this by considering different sizes of claim. From the insurer's point of view, the amount it has to pay out on small claims (those that are nowhere near the retention limit) will increase by k . However, the amount paid on claims that were already above the limit will not increase at all (and the amount paid on claims that didn't reach the limit before but now do will increase, but by less than k).

The reinsurer's mean claim amount will increase by a factor of more than k to compensate.

A similar approach can also be taken in situations where the retention limit is linked to some index of inflation.

Of course if the retention limit increases by a factor of k as well, both mean claim amounts (for the insurer and reinsurer) will increase by the same factor.

When examining the details of a reinsurance arrangement in real life it is very important to check whether the retention limits are fixed or are linked to an agreed inflation index. There are special published indices specifically for use in connection with general insurance claims.



The payments of the insurer, y , and the reinsurer, z , with retention M and inflation factor k would be:

```
y <- pmin(k*x,M)
z <- pmax(0,k*x-M)
```

We can then calculate moments, probabilities and quantiles as before.

4 Estimation

Consider the problem of estimation in the presence of excess of loss reinsurance. Suppose that the claims record shows only the net claims paid by the insurer. A typical claims record might be:

$$x_1, x_2, M, x_3, M, x_4, x_5, \dots \quad (18.6)$$

and an estimate of the underlying gross claims distribution is required.

As before, we wish to estimate the parameters for the distribution we have assumed for the claims.

The method of moments is not available since even the mean claim amount cannot be computed. On the other hand, it may be possible to use the method of percentiles without alteration; this would happen if the retention level M is high and only the higher sample percentiles were affected by the (few) reinsurance claims.

The statistical terminology for a sample of the form (18.6) is censored. In general, a censored sample occurs when some values are recorded exactly and the remaining values are known only to exceed a particular value, here the retention level M .

Maximum likelihood can be applied to censored samples. The likelihood function is made up of two parts. If the values of $x_1, x_2, \dots, x_n$ are recorded exactly these contribute a factor of:

$$L_1(\theta) = \prod_{i=1}^n f(x_i; \theta)$$

If a further m claims are referred to the reinsurer, then the insurer records a payment of M for each of these claims. These censored values then contribute a factor:

$$L_2(\theta) = \prod_{j=1}^m P(X > M) \quad \text{ie } [P(X > M)]^m$$

The complete likelihood function is:

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta) \times [1 - F(M; \theta)]^m$$

where $F(\cdot; \theta)$ is the CDF of the claims distribution.

The reason for multiplying is that the likelihood reflects the probability of getting the n claims with known values and m claims exceeding M . Also, we are assuming that the claims are independent.



In R, we can define the censored log-likelihood function and use the function `nlm` on the negative value of this as before.



Question

Claims from a portfolio are believed to follow an $Exp(\lambda)$ distribution. The insurer has effected individual excess of loss reinsurance with a retention limit of 1,000.

The insurer observes a random sample of 100 claims, and finds that the average amount of the 90 claims that do not exceed 1,000 is 82.9. There are 10 claims that do exceed the retention limit.

Calculate the maximum likelihood estimate of the parameter λ .

Solution

Here $X \sim Exp(\lambda)$ and $P(X > 1,000) = e^{-1,000\lambda}$. So the likelihood function is:

$$L(\lambda) = \lambda e^{-\lambda x_1} \lambda e^{-\lambda x_2} \dots \lambda e^{-\lambda x_{90}} \times (e^{-1,000\lambda})^{10} = \lambda^{90} e^{-(10,000 + \sum x_i)\lambda}$$

Taking logs:

$$\ln L = 90 \ln \lambda - (10,000 + \sum x_i)\lambda$$

Differentiating with respect to λ :

$$\frac{\partial}{\partial \lambda} \ln L = \frac{90}{\lambda} - (10,000 + \sum x_i)$$

This is equal to 0 when:

$$\lambda = \frac{90}{10,000 + \sum x_i} = \frac{90}{10,000 + (90 \times 82.9)} = 0.005154$$

Differentiating again:

$$\frac{\partial^2}{\partial \lambda^2} \ln L = -\frac{90}{\lambda^2}$$

This is negative when $\lambda = 0.005154$. (In fact it is always negative.) So we have a maximum turning point and hence $\hat{\lambda} = 0.005154$.

5 Policy excess

Insurance policies with an excess are common in motor insurance and many other kinds of property and accident insurance. Under this kind of policy, the insured agrees to carry the full burden of the loss up to a limit, L , called the excess. If the loss is an amount X , greater than L , then the policyholder will claim only $X - L$. If Y is the amount actually paid by the insurer, then:

$$Y = 0 \quad \text{if } X \leq L$$

$$Y = X - L \quad \text{if } X > L$$

Clearly, the premium due on any policy with an excess will be less than that on a policy without an excess.

This assumes that some of the saving is actually passed on to the policyholder. A policy excess may also be referred to as a *deductible*.

The position of the insurer for a policy with an excess is exactly the same as that of the reinsurer under excess of loss reinsurance. The position of the policyholder as far as losses are concerned is exactly the same as that of an insurer with an excess of loss reinsurance contract.

In practice, expenses form a significant part of the insurance cost. So the presence of an excess might not affect the premium as much as might be expected. A premium calculated ignoring expenses is called a 'risk premium'.



Question

An insurer believes that claims from a particular type of policy follow a Pareto distribution with parameters $\alpha = 2$ and $\lambda = 900$. The insurer wishes to introduce a policy excess so that 20% of losses result in no claim to the insurer.

Calculate the size of the excess.

Solution

Let L be the size of the excess. The insurer wants to set L so that $P(X < L) = 0.2$. Using the given loss distribution, we have:

$$P(X < L) = 1 - \left(\frac{900}{900 + L} \right)^2$$

So we require:

$$1 - \left(\frac{900}{900 + L} \right)^2 = 0.2$$

Rearranging:

$$\left(\frac{900}{900+L} \right)^2 = 0.8$$

$$\Rightarrow \frac{900}{900+L} = \sqrt{0.8}$$

$$\Rightarrow 900+L = \frac{900}{\sqrt{0.8}}$$

$$\Rightarrow L = \frac{900}{\sqrt{0.8}} - 900 = 106.23$$

The chapter summary starts on the next page so that you can keep all the chapter summaries together for revision purposes.

Chapter 18 Summary

Reinsurance

Reinsurance is insurance for insurance companies. By using reinsurance, the insurer seeks to protect itself from large claims. The mean amount paid by the insurer is reduced, and the variance of the amount paid by the insurer is reduced.

Reinsurance may be proportional or non-proportional (*ie* excess of loss).

We use the following notation:

X is the gross claim amount random variable

Y is the net claim amount, *ie* the amount of the claim paid by the insurer

Z is the amount paid by the reinsurer

Proportional reinsurance

Under proportional reinsurance, the insurer and the reinsurer split the claim in pre-defined proportions. For a claim amount X , the amount paid by the insurer is $Y = \alpha X$ and the amount paid by the reinsurer is $Z = (1 - \alpha)X$ where α is known as the retained proportion or retention level, $0 < \alpha < 1$.

$$E(Y) = \alpha E(X)$$

$$E(Z) = (1 - \alpha)E(X)$$

$$\text{var}(Y) = \alpha^2 \text{var}(X)$$

$$\text{var}(Z) = (1 - \alpha)^2 \text{var}(X)$$

Non-proportional reinsurance (individual excess of loss)

Under individual excess of loss, the insurer will pay any claim in full up to an amount M , the retention level. Any amount above M will be met by the reinsurer.

$$Y = \begin{cases} X & \text{if } X \leq M \\ M & \text{if } X > M \end{cases}$$

$$Z = \begin{cases} 0 & \text{if } X \leq M \\ X - M & \text{if } X > M \end{cases}$$

$$E(Y) = \int_0^M x f_X(x) dx + \int_M^\infty M f_X(x) dx$$

$$E(Z) = \int_M^\infty (x - M) f_X(x) dx$$

$$E(Y^2) = \int_0^M x^2 f_X(x) dx + \int_M^\infty M^2 f_X(x) dx$$

$$E(Z^2) = \int_M^\infty (x - M)^2 f_X(x) dx$$

Reinsurer's conditional claims distribution

It may be the case that the reinsurer is only informed of claims greater than the retention level M . In this case, the reinsurer observes claims from a truncated (or conditional) distribution. Let W be the random variable associated with this distribution, then:

$$W = Z | Z > 0 = X - M | X > M$$

$$f_W(w) = \frac{f_X(w+M)}{P(X > M)} = \frac{f_X(w+M)}{P(Z > 0)}$$

$$E(W) = \frac{E(Z)}{P(Z > 0)}$$

Excesses

When a policy excess applies, the policyholder pays for the first part of each loss up to an excess level L . Any amount greater than L will be met by the insurer. The positions of the policyholder and the insurer as far as losses are concerned are the same as those of the insurer and the reinsurer respectively under individual excess of loss reinsurance. When a policy excess applies, the insurer's conditional distribution takes the same form as that of the reinsurer's conditional distribution above.

Inflation and individual excess of loss reinsurance

If claims are inflated by a factor of k but the retention level remains fixed at M then the amount paid by the insurer is:

$$Y = \begin{cases} kX & \text{if } X \leq \frac{M}{k} \\ M & \text{if } X > \frac{M}{k} \end{cases}$$

The amount paid by the reinsurer is:

$$Z = \begin{cases} 0 & \text{if } X \leq \frac{M}{k} \\ kX - M & \text{if } X > \frac{M}{k} \end{cases}$$

Estimation of parameters from a censored sample

The likelihood function of a vector of parameters $\underline{\theta}$, based on a sample of n exact observations and m censored observations known to exceed M is:

$$L(\underline{\theta}) = \left[\prod_{i=1}^n f_X(x_i) \right] [P(X > M)]^m$$

assuming that the observations are realisations of $n + m$ IID random variables.



Chapter 18 Practice Questions

- 18.1 An insurer insures a risk for which individual claim sizes (in £000s) have mean 500 and standard deviation 250. The insurer arranges excess of loss reinsurance for this risk with a retention limit of £1,000,000.

Calculate the proportion of claims from this risk for which the insurer expects to receive a payment from the reinsurer if the loss distribution is:

- (a) gamma
- (b) lognormal.

- 18.2 Claims arising from a particular portfolio have a Pareto distribution with parameters $\alpha = 6$ and $\lambda = 200$. The insurer effects individual excess of loss reinsurance with a retention limit of 80.

- (i) Calculate the insurer's expected claim amount before and after reinsurance.
- (ii) Calculate the mean amount paid by the reinsurer on claims in which it is involved.

- 18.3 A sample of a reinsurer's payments made under a proportional reinsurance arrangement consists of the following values, in units of thousands of pounds:

4.6, 6.8, 22.9, 1.4, 3.8, 10.2, 19.4, 32.1

If the original claim amounts have a $\text{Gamma}(\alpha, \lambda)$ distribution, and the retained proportion is 80%, determine the distribution of the reinsurer's claim payments. Hence estimate the parameters α and λ using the method of moments.

- 18.4 If $X \sim \log N(7.5, 0.85^2)$, calculate:

(a)
$$\int_{1,000}^{5,000} f(x) dx$$

(b)
$$\int_0^{1,000} x f(x) dx$$

(c)
$$\int_{5,000}^{\infty} x^2 f(x) dx$$

- 18.5 Claims from a portfolio are believed to have a Pareto distribution with parameters α and λ . In Year 0, $\alpha = 6$ and $\lambda = 1,000$. An excess of loss reinsurance arrangement is in force, with a retention limit of 500. Inflation is a constant 10% *pa*.

- (i) Determine the distribution of the gross claim amounts in Years 1 and 2.
- (ii) Calculate the reinsurer's mean claim payment on all claims in Years 0, 1 and 2.

18.6 Claim amounts from a portfolio follow a Weibull distribution with PDF:

$$f(x) = 2cx e^{-cx^2}, x \geq 0$$

An individual excess of loss reinsurance arrangement with retention limit $M=3$ is in force. A sample of the *reinsurer's* non-zero payment amounts gives the following values:

$$n = 10 \quad \sum w_i = 8.7 \quad \sum w_i^2 = 92.3$$

where the units are millions of pounds. Calculate the maximum likelihood estimate of c .

18.7 (i) A random variable X has the lognormal distribution with density function $f(x)$ and parameters μ and σ . Show that for $a > 0$:

Exam style

$$\int_a^\infty x f(x) dx = \exp\left(\mu + \frac{\sigma^2}{2}\right) \left(1 - \Phi\left(\frac{\log a - \mu - \sigma^2}{\sigma}\right)\right)$$

where Φ is the cumulative distribution function of the standard normal distribution. [4]

(ii) Claims under a particular class of insurance follow a lognormal distribution with mean 9.070 and standard deviation of 10.132 (figures in £000s). In any one year 20% of policies are expected to give rise to a claim.

An insurance company has 200 policies on its books and wishes to take out individual excess of loss reinsurance to cover all the policies in the portfolio. The reinsurer has quoted premiums for two levels of reinsurance as follows (figures in £000s):

Retention limit	Premium
25	50
30	40

- Calculate the probability, under each reinsurance arrangement, that a claim arising will involve the reinsurer.
- By investigating the average amount of each claim ceded to the reinsurer, calculate which of the retention levels gives the best value for money for the insurer (ignoring the insurer's attitude to risk).
- The following year, assuming all other things equal, the insurer believes that inflation will increase the mean and standard deviation of the claims in its portfolio by 8%. If the reinsurer charges the same premiums as before, determine which of the retention levels will give best value for money next year. [18]

[Total 22]

18.8

Exam style

- (i) Loss amounts from a particular type of insurance have a Pareto distribution with parameters α and λ . If the company applies a policy excess, E , derive the distribution function of claim amounts paid by the insurer. [3]
- (ii) Assuming that $\alpha = 4$ and $\lambda = 15$, calculate the mean claim amount paid by the insurer:
- (a) with no policy excess ($E = 0$),
- (b) with an excess of 10 ($E = 10$). [2]
- (iii) Using your answers to (ii), comment on the effect of introducing a policy excess. [2]
- [Total 7]

18.9

Exam style

Losses from a group of travel insurance policies are assumed to follow a Pareto distribution with parameters $\alpha = 4.5$ and $\lambda = 3,000$.

Next year losses are expected to increase by 3%, and the insurer has decided to introduce a policy excess of 100 per claim.

Calculate the probability that a loss next year is borne entirely by the policyholder. [2]

The solutions start on the next page so that you can
separate the questions and solutions.



Chapter 18 Solutions

18.1 (a) *Gamma distribution*

Let X denote the loss random variable and suppose that $X \sim \text{Gamma}(\alpha, \lambda)$. Then:

$$E(X) = \frac{\alpha}{\lambda} = 500 \quad \text{and} \quad \text{var}(X) = \frac{\alpha}{\lambda^2} = 250^2$$

Solving these equations simultaneously gives:

$$\alpha = 4 \quad \text{and} \quad \lambda = 0.008$$

The reinsurer will make a payment if the claim size exceeds £1m. Since we are working in £000s, we have to calculate:

$$P(X > 1,000)$$

To do this, we can use the relationship between the gamma and chi-squared distributions:

$$X \sim \text{Gamma}(\alpha, \lambda) \Rightarrow 2\lambda X \sim \chi^2_{2\alpha}$$

So:

$$P(X > 1,000) = P(2\lambda X > 2,000\lambda) = P(\chi^2_8 > 16) = 1 - 0.9576 = 0.0424$$

(b) *Lognormal distribution*

If $X \sim \log N(\mu, \sigma^2)$, then:

$$E(X) = e^{\mu + \frac{1}{2}\sigma^2} = 500 \quad \text{and} \quad \text{var}(X) = e^{2\mu + \sigma^2} (e^{\sigma^2} - 1) = 250^2$$

Squaring the first equation and substituting this into the second gives:

$$e^{\sigma^2} - 1 = \frac{250^2}{500^2} = 0.25 \Rightarrow \sigma^2 = \ln 1.25 = 0.22314$$

Then, from the equation for $E(X)$ we have:

$$\mu = \ln 500 - \frac{1}{2}\sigma^2 = 6.1030$$

and:

$$\begin{aligned} P(X > 1,000) &= P\left(Z > \frac{\ln 1,000 - 6.103036}{\sqrt{0.2231436}}\right) = P(Z > 1.70354) \\ &= 1 - 0.95576 = 0.04424 \end{aligned}$$

18.2 (i) **Insurer's expected claim payments**

Let X denote the gross claim amount random variable. Then $X \sim Pa(6, 200)$ and:

$$E(X) = \frac{200}{6-1} = 40$$

ie the mean claim amount before reinsurance is 40.

The insurer's net claim amount is:

$$Y = \begin{cases} X & \text{if } X \leq 80 \\ 80 & \text{if } X > 80 \end{cases}$$

So:

$$E(Y) = \int_0^{80} x \frac{6(200)^6}{(200+x)^7} dx + 80P(X > 80)$$

The integral can be evaluated by substitution. Let $u = 200 + x$. Then:

$$\begin{aligned} \int_0^{80} x \frac{6(200)^6}{(200+x)^7} dx &= \int_{200}^{280} (u-200) \frac{6(200)^6}{u^7} du \\ &= 6(200)^6 \int_{200}^{280} (u^{-6} - 200u^{-7}) du \\ &= 6(200)^6 \left[\frac{u^{-5}}{-5} - \frac{200u^{-6}}{-6} \right]_{200}^{280} \\ &= 6(200)^6 \left[\left(-\frac{280^{-5}}{5} + \frac{200 \times 280^{-6}}{6} \right) - \left(-\frac{200^{-5}}{5} + \frac{200 \times 200^{-6}}{6} \right) \right] \\ &= 21.9378 \end{aligned}$$

Alternatively, we could integrate by parts.

Also:

$$P(X > 80) = 1 - F(80) = \left(\frac{200}{200+80} \right)^6 = 0.1328$$

So the mean claim amount after reinsurance is:

$$E(Y) = 21.9378 + 80 \times 0.1328 = 32.5626$$

(ii) **Mean amount paid by the reinsurer on claims in which it is involved**

The mean amount paid by the reinsurer on all claims (including those where the reinsurer makes no payment) is:

$$E(Z) = E(X) - E(Y) = 40 - 32.5626 = 7.4374$$

and the mean amount paid by the reinsurer on claims in which it is involved is:

$$\frac{E(Z)}{P(X > 80)} = \frac{7.4374}{0.1328} = 56$$

Alternatively, we could say that W , the reinsurer's conditional claim payment random variable has a $Pa(6, 280)$ distribution. The mean of this distribution is $\frac{280}{6-1} = 56$.

18.3 Let X be the gross claim amount random variable and Z be the reinsurer's claim payment. Then $X \sim \text{Gamma}(\alpha, \lambda)$ and $Z = 0.2X$. The moment generating function of Z is:

$$M_Z(t) = E(e^{tZ}) = E(e^{0.2tX}) = M_X(0.2t) = \left(1 - \frac{0.2t}{\lambda}\right)^{-\alpha} = \left(1 - \frac{t}{5\lambda}\right)^{-\alpha}, \quad t < 5\lambda$$

This is the MGF of the $\text{Gamma}(\alpha, 5\lambda)$ distribution. By the uniqueness property of MGFs, it follows that $Z \sim \text{Gamma}(\alpha, 5\lambda)$. So:

$$E(Z) = \frac{\alpha}{5\lambda} \quad \text{and} \quad \text{var}(Z) = \frac{\alpha}{(5\lambda)^2} = \frac{\alpha}{25\lambda^2}$$

The sample mean and n -denominator variance are:

$$\bar{z} = \frac{\sum z_i}{8} = \frac{101.2}{8} = 12.65$$

and:

$$s^2 = \frac{\sum z_i^2}{8} - \bar{z}^2 = \frac{2,119.02}{8} - 12.65^2 = 104.855$$

The method of moments estimates of α and λ are the solutions of the equations:

$$\frac{\alpha}{5\lambda} = 12.65 \quad \text{and} \quad \frac{\alpha}{25\lambda^2} = 104.855$$

Solving these gives $\hat{\alpha} = 1.526$ and $\hat{\lambda} = 0.02413$.

- 18.4 (a) Using the truncated moments formula with $k = 0$:

$$\begin{aligned}\int_{1,000}^{5,000} f(x) dx &= \Phi\left(\frac{\ln 5,000 - 7.5}{0.85}\right) - \Phi\left(\frac{\ln 1,000 - 7.5}{0.85}\right) \\ &= \Phi(1.19670) - \Phi(-0.69676) \\ &= 0.88429 - 0.24298 \\ &= 0.6413\end{aligned}$$

This is $P(1,000 < X < 5,000)$.

- (b) Using the formula with $k = 1$:

$$\begin{aligned}\int_0^{1,000} x f(x) dx &= e^{7.5 + \frac{1}{2} \times 0.85^2} \left[\Phi\left(\frac{\ln 1,000 - 7.5}{0.85} - 0.85\right) - \Phi(-\infty) \right] \\ &= e^{7.86125} [\Phi(-1.54676) - 0] \\ &= 2,594.76 \times 0.06096 \\ &= 158.2\end{aligned}$$

- (c) Using the formula with $k = 2$:

$$\begin{aligned}\int_{5,000}^{\infty} x^2 f(x) dx &= e^{2(7.5) + 2 \times 0.85^2} \left[\Phi(\infty) - \Phi\left(\frac{\ln 5,000 - 7.5}{0.85} - 2(0.85)\right) \right] \\ &= e^{16.445} [1 - \Phi(-0.50330)] \\ &= 13,866,688 (1 - 0.30738) \\ &= 9.604m\end{aligned}$$

- 18.5 (i) ***Distribution of insurer's claim payments before reinsurance***

Let X_j be the gross claim amount random variable in Year j . Then $X_0 \sim Pa(6, 1000)$ and $X_1 = 1.1X_0$. The Pareto distribution does not have a moment generating function, but we can determine the distribution of X_1 by considering its CDF. For $x > 0$:

$$F_{X_1}(x) = P(X_1 \leq x) = P(1.1X_0 \leq x) = P\left(X_0 \leq \frac{x}{1.1}\right) = F_{X_0}\left(\frac{x}{1.1}\right) = 1 - \left(\frac{1,000}{1,000 + \frac{x}{1.1}}\right)^6$$

Multiplying the numerator and the denominator of the bracketed fraction by 1.1, we see that:

$$F_{X_1}(x) = 1 - \left(\frac{1,100}{1,100 + x}\right)^6, \quad x > 0$$

This is the CDF of the $Pa(6, 1100)$ distribution. So $X_1 \sim Pa(6, 1100)$.

Inflation has no effect on the first parameter, but the second parameter has increased by 10%.

Similarly, $X_2 \sim Pa(6, 1210)$.

(ii) **Reinsurer's expected claim payments**

Let Z_j be the reinsurer's claim payment random variable in Year j . Then:

$$E(Z_0) = \int_{500}^{\infty} (x - 500) f_{X_0}(x) dx = \int_{500}^{\infty} (x - 500) \frac{6(1,000)^6}{(1,000 + x)^7} dx$$

Substituting $t = x - 500$, we see that:

$$E(Z_0) = \int_0^{\infty} t \frac{6(1,000)^6}{(1,500 + t)^7} dt$$

We can rewrite this as follows:

$$E(Z_0) = \left(\frac{1,000}{1,500} \right)^6 \int_0^{\infty} t \frac{6(1,500)^6}{(1,500 + t)^7} dt$$

The integrand is of the form $t f_T(t)$, where $T \sim Pa(6, 1500)$. So:

$$E(Z_0) = \left(\frac{1,000}{1,500} \right)^6 E(T) = \left(\frac{1,000}{1,500} \right)^6 \times \frac{1,500}{5} = 26.337$$

The only change from Year 0 to Year 1 is in the λ parameter. So, using the same approach:

$$E(Z_1) = \int_{500}^{\infty} (x - 500) \frac{6(1,100)^6}{(1,100 + x)^7} dx = \int_0^{\infty} t \frac{6(1,100)^6}{(1,600 + t)^7} dt = \left(\frac{1,100}{1,600} \right)^6 \int_0^{\infty} t \frac{6(1,600)^6}{(1,600 + t)^7} dt$$

The final integral is the mean of the $Pa(6, 1600)$ distribution. So:

$$E(Z_1) = \left(\frac{1,100}{1,600} \right)^6 \times \frac{1,600}{5} = 33.790$$

Similarly:

$$\begin{aligned} E(Z_2) &= \int_{500}^{\infty} (x - 500) \frac{6(1,210)^6}{(1,210 + x)^7} dx = \int_0^{\infty} t \frac{6(1,210)^6}{(1,710 + t)^7} dt = \left(\frac{1,210}{1,710} \right)^6 \int_0^{\infty} t \frac{6(1,710)^6}{(1,710 + t)^7} dt \\ &= \left(\frac{1,210}{1,710} \right)^6 \times \frac{1,710}{5} = 42.930 \end{aligned}$$

The percentage increase from Year 0 to Year 1 is 28.3%, and the percentage increase from Year 1 to Year 2 is 27.1%. These figures are more than 10% as expected.

- 18.6 Since we are given information about claim payments made by the reinsurer, we need to consider the reinsurer's conditional claim amount random variable. This has PDF:

$$g(w) = \frac{f(w+M)}{1-F(M)}$$

The gross claim amount random variable has a Weibull distribution with parameters c and 2. So:

$$g(w) = \frac{2c(w+M)e^{-c(w+M)^2}}{e^{-cM^2}} = 2c(w+3)e^{-c(w^2+6w)}$$

So the likelihood function based on a random sample of n payments made by the reinsurer is:

$$L(c) = 2^n c^n \prod_{i=1}^n (w_i + 3) \exp \left(-c \sum_{i=1}^n (w_i^2 + 6w_i) \right)$$

Taking logs:

$$\ln L = n \ln 2 + n \ln c + \sum_{i=1}^n \ln(w_i + 3) - c \sum_{i=1}^n (w_i^2 + 6w_i)$$

Differentiating with respect to c :

$$\frac{\partial}{\partial c} \ln L = \frac{n}{c} - \sum_{i=1}^n (w_i^2 + 6w_i)$$

This is 0 when:

$$c = \frac{n}{\sum_{i=1}^n (w_i^2 + 6w_i)}$$

Differentiating again:

$$\frac{\partial^2}{\partial c^2} \ln L = -\frac{n}{c^2}$$

This is negative, so we have a maximum.

Substituting in the given numerical values, we find that:

$$\hat{c} = \frac{10}{92.3 + 6 \times 8.7} = 0.0692$$

18.7 (i) **Proof**

We want to simplify the integral:

$$\int_a^{\infty} x \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{\log x - \mu}{\sigma}\right)^2} dx$$

Making the substitution $u = \frac{\log x - \mu}{\sigma} - \sigma$, the integral becomes:

$$\int_{\frac{\log a - \mu}{\sigma} - \sigma}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}(u+\sigma)^2} \sigma e^{\mu+u\sigma+\sigma^2} du \quad [1]$$

Multiplying out the brackets in the exponent and simplifying, we have:

$$e^{\mu+\frac{1}{2}\sigma^2} \int_{\frac{\log a - \mu}{\sigma} - \sigma}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}u^2} du \quad [1]$$

This integrand is the PDF of the standard normal distribution and the integral is:

$$P\left(N(0,1) > \frac{\log a - \mu}{\sigma} - \sigma\right) \quad [1]$$

So we have:

$$\int_a^{\infty} x f(x) dx = e^{\mu+\frac{1}{2}\sigma^2} \left[1 - \Phi\left(\frac{\log a - \mu}{\sigma} - \sigma\right)\right] \quad [1]$$

This is the required result.

(ii)(a) **Probability**

We first need the parameter values for the lognormal distribution. Using the formulae for the mean and variance of the lognormal distribution from page 14 of the *Tables* we have the following equations:

$$e^{\mu+\frac{1}{2}\sigma^2} = 9.070 \quad \text{and} \quad e^{2\mu+\sigma^2}(e^{\sigma^2} - 1) = 10.132^2$$

Solving these simultaneous equations (by squaring the first equation and then substituting into the second equation), we obtain the values:

$$\sigma^2 = 0.80999 \quad \text{and} \quad \mu = 1.79998 \quad [2]$$

The probability that a claim involves the reinsurer is the probability that it exceeds the retention limit. So if X represents the amount of a claim, we have, for the first reinsurance arrangement:

$$\begin{aligned} P(X > 25) &= P(\log N(\mu, \sigma^2) > 25) = P(N(\mu, \sigma^2) > \log 25) \\ &= P\left(N(0, 1) > \frac{\log 25 - \mu}{\sigma}\right) = 1 - \Phi\left(\frac{\log 25 - \mu}{\sigma}\right) \end{aligned} \quad [1]$$

Substituting in the values for μ and σ^2 , we get:

$$P(X > 25) = 1 - \Phi(1.57656) = 0.05745 \quad [1]$$

So the probability that a claim will involve the reinsurer if the first arrangement is in force is 5.745%.

Using exactly the same argument for the second arrangement, we get:

$$P(X > 30) = 1 - \Phi\left(\frac{\log 30 - \mu}{\sigma}\right) = 1 - \Phi(1.77914) = 0.03761 \quad [1]$$

So the probability that a claim will involve the reinsurer if the second arrangement is in force is 3.761%.

(ii)(b) **Better arrangement**

Consider the first arrangement. The amount ceded to the reinsurer (*ie* the amount paid by the reinsurer on a claim) is:

$$Z = \begin{cases} 0 & \text{if } X \leq 25 \\ X - 25 & \text{if } X > 25 \end{cases}$$

So:

$$E(Z) = \int_{25}^{\infty} (x - 25) f(x) dx = \int_{25}^{\infty} x f(x) dx - 25 \int_{25}^{\infty} f(x) dx \quad [1]$$

where $f(x)$ is the PDF of the original lognormal distribution.

We can calculate the first of these integrals by using the result from the first part of the question:

$$\begin{aligned} \int_{25}^{\infty} x f(x) dx &= e^{\mu + \frac{1}{2}\sigma^2} \left[1 - \Phi\left(\frac{\log 25 - \mu - \sigma^2}{\sigma}\right) \right] \\ &= 9.070 [1 - \Phi(0.67657)] \\ &= 9.070 \times 0.24934 \\ &= 2.2615 \end{aligned} \quad [1]$$

The second integral is just the probability that we worked out in part (ii)(a). So:

$$E(Z) = 2.2615 - 25 \times 0.05745 = 0.8253 \quad [1]$$

Working in exactly the same way for the second arrangement (where Z is now the amount paid by the reinsurer in excess of 30), we have:

$$\begin{aligned} E(Z) &= 9.070[1 - \Phi(0.87915)] - 30 \times 0.03761 \\ &= 9.070 \times 0.18966 - 30 \times 0.03761 \\ &= 0.5919 \end{aligned} \quad [1]$$

So the expected amount paid out by the reinsurer per £1 of premium is (under the first arrangement):

$$\frac{200 \times 0.2 \times 0.8253}{50} = \text{£}0.660 \quad [1]$$

Under the second arrangement, it is:

$$\frac{200 \times 0.2 \times 0.5919}{40} = \text{£}0.592 \quad [1]$$

So, other things being equal, the first arrangement looks better value.

(ii)(c) **Better arrangement under new circumstances**

The new mean and standard deviation are now 9.7956 and 10.94256 respectively. So we can calculate the new parameter values:

$$e^{\mu + \frac{1}{2}\sigma^2} = 9.7956 \quad \text{and} \quad e^{2\mu + \sigma^2}(e^{\sigma^2} - 1) = 10.94256^2$$

Solving these exactly as we did before, we find that $\mu = 1.87694$ and σ^2 is unchanged at 0.80999. [2]

So the value for $E(Z)$ is now (under the first arrangement):

$$\begin{aligned} E(Z) &= 9.7956 \left[1 - \Phi \left(\frac{\log 25 - 1.87694 - 0.80999}{\sqrt{0.80999}} \right) \right] \\ &\quad - 25 \times \left[1 - \Phi \left(\frac{\log 25 - 1.87694}{\sqrt{0.80999}} \right) \right] \\ &= 9.7956[1 - \Phi(0.59106)] - 25[1 - \Phi(1.49105)] \\ &= 9.7956 \times 0.27724 - 25 \times 0.06797 \\ &= 1.0164 \end{aligned} \quad [2]$$

Under the second arrangement:

$$\begin{aligned}
 E(Z) &= 9.7956 \left[1 - \Phi \left(\frac{\log 30 - 1.87694 - 0.80999}{\sqrt{0.80999}} \right) \right] \\
 &\quad - 30 \times \left[1 - \Phi \left(\frac{\log 30 - 1.87694}{\sqrt{0.80999}} \right) \right] \\
 &= 9.7956 [1 - \Phi(0.79364)] - 30 [1 - \Phi(1.69363)] \\
 &= 9.7956 \times 0.21370 - 30 \times 0.04517 \\
 &= 0.73830 \quad [2]
 \end{aligned}$$

So working exactly as before, the payment per £1 of premium under the first arrangement is now:

$$\frac{200 \times 0.2 \times 1.0164}{50} = \text{£}0.813$$

In the second arrangement the corresponding figure is:

$$\frac{200 \times 0.2 \times 0.73830}{40} = \text{£}0.738$$

So the first arrangement is still better value for the insurer. [1]

18.8 (i) **Distribution function**

Let X denote the amount of the loss and Y denote the amount paid by the insurer in respect of the loss. With a policy excess of E in force, we have:

$$Y = X - E \mid X > E$$

The CDF of Y is given by:

$$\begin{aligned}
 F_Y(y) &= P(Y \leq y) = P(X - E \leq y \mid X > E) \\
 &= \frac{P(X - E \leq y \text{ and } X > E)}{P(X > E)} \\
 &= \frac{P(X \leq y + E \text{ and } X > E)}{P(X > E)} \\
 &= \frac{P(E < X \leq y + E)}{P(X > E)} \\
 &= \frac{F_X(y + E) - F_X(E)}{1 - F_X(E)} \quad [2]
 \end{aligned}$$

Since $X \sim Pa(\alpha, \lambda)$:

$$\begin{aligned}
 F_Y(y) &= \frac{\left[1 - \left(\frac{\lambda}{\lambda + y + E}\right)^\alpha\right] - \left[1 - \left(\frac{\lambda}{\lambda + E}\right)^\alpha\right]}{1 - \left[1 - \left(\frac{\lambda}{\lambda + E}\right)^\alpha\right]} \\
 &= \frac{\left(\frac{\lambda}{\lambda + E}\right)^\alpha - \left(\frac{\lambda}{\lambda + y + E}\right)^\alpha}{\left(\frac{\lambda}{\lambda + E}\right)^\alpha} \\
 &= 1 - \left(\frac{\lambda + E}{\lambda + y + E}\right)^\alpha, \quad y > 0
 \end{aligned} \tag{1}$$

So $Y \sim Pa(\alpha, \lambda + E)$.

(ii) **Mean values**

The mean of the $Pa(\alpha, \lambda + E)$ distribution is $\frac{\lambda + E}{\alpha - 1}$.

If $E = 0$ then $E(Y) = \frac{15}{3} = 5$. [1]

If $E = 10$ then $E(Y) = \frac{25}{3} = 8\frac{1}{3}$. [1]

(iii) **Effect of introducing a policy excess**

Introducing a policy excess of E increases the mean claim amount paid by the insurer by $\frac{E}{\alpha - 1}$.

This is because small losses are met in full by the policyholder. [1]

It may still be advantageous to the insurer to introduce a policy excess, since although the average claim payment will increase, fewer claim payments will be made.

18.9 Let X be the loss amount random variable for this year, and let X' be the loss amount random variable for next year. Then:

$$X \sim Pa(4.5, 3000) \quad \text{and} \quad X' = 1.03X$$

The probability that a loss next year is borne entirely by the policyholder is:

$$P(X' \leq 100) = P\left(X \leq \frac{100}{1.03}\right) = 1 - \left(\frac{3,000}{3,000 + \frac{100}{1.03}}\right)^{4.5} = 0.13353 \tag{2}$$

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